Integrating Scanner Data into CPI Production: Institutional and operational considerations

UN-CEBD Task Team
on Scanner Data

Framing the challenge

Integrating scanner data into CPI production is a progression of decisions from strategic motivation through to sustainable production.

1. Motivation & trade-offs

Why are we interested in this data and what are the opportunities and constraints?

why and what are the constraints

2. Decisions

Should we use this data, and how should we use it within the CPI?

whether and how to use data

3. Readiness

Do we have the capability, systems, governance and resources in place?

capability and resources

4. Implementation

How will we acquire, process and integrate the data into CPI production?

systems and processes

5. Sustain

Can we produce reliable CPI outputs over time and manage risks effectively?

production and risk

From strategic motivation to sustainable CPI production

Successful integration requires coordinated decisions, investment and ongoing management across all stages.

Motivations and trade-offs

Why are we interested in these data and what are the opportunities and constraints?

Strategic motivations

- Improve timeliness and frequency
- Increase granularity and analytical potential
- Support methodological development
- Improve efficiency and sustainability
- Enhance coverage of rapidly changing markets
- Support wider statistical use (weights, QA, other outputs)

Motivations shape design

Different priorities can lead to different implementation choices:

Motivation	Likely design implication
Coverage / representativity	Focus on high-coverage sources; dynamic or expanded product universe may be more attractive
Timeliness / frequency	Greater need for high-frequency pipelines, automation and rapid QA
Efficiency	May prioritise replacing or streamlining existing collection processes
Continuity / stability	More cautious introduction, sampled/static or hybrid approaches may be preferred
Analytical use	Data may initially support QA, weights or supplementary analysis before direct compilation

Opportunities and constraints

Tension	What it means in practice
Coverage vs representativity	High data volume does not guarantee full market coverage
Detail vs complexity	Rich, granular data increases processing and QA burden
Timeliness vs stability	Faster data may introduce volatility or revision challenges
Efficiency vs dependency	Reduced collection burden increases reliance on external providers
Innovation vs continuity	New methods may improve measurement but risk breaking series

Decisions

Should we use these data and how?

Guiding principles

Conceptual consistency with CPI framework

Transparency and reproducibility

Confidentiality and ethical use

Data quality and suitability

Continuity and comparability over time

Should we use the data?

Does the data align with CPI scope and concepts?

Does it improve quality or coverage?

Is access sustainable over time?

Are risks and limitations understood?

How should we use the data?

Replace or supplement existing data sources?

Static (sampled) or dynamic (full dataset) approach?

Where and how does it enter the CPI?

Full integration or targeted use?

"How" to start?

- Address problems with a reduced degree of complexity first;
- As maturity and expertise is developed, increase the complexity.

- Consider a reduced number of sectors and retailers first.
- Increase complexity as problems with the initial ones are solved and expertise gained.

Readiness

Do we have the capability, governance and systems in place?

Resource implications

Integration requirements

New data
Legal and data access considerations, potential acquisition costs and ongoing monitoring

New elementary indices
Incorporation of new elementary índices formulations & strata

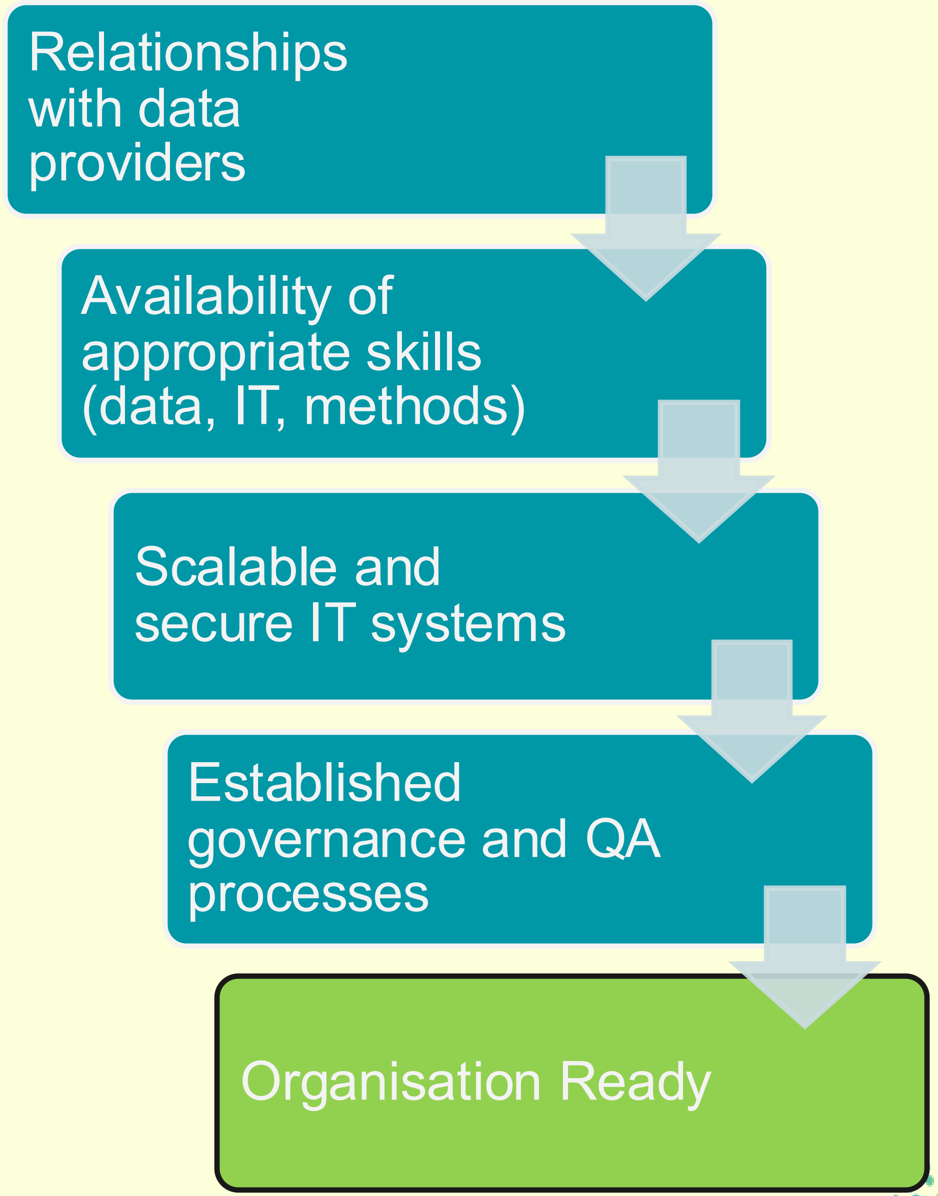
IT system changes
Changes in the CPI systems for the new calculations required

Staff routine
Changes in the CPI analysis routines, new training & skill requirements.

Dissemination
Evaluation of impact of changes when linking old and new series and Communication of changes to users Potential requirement for parallel runs & impact analyses prior to production

Significant extra funding is required

Organisational readiness



Implementation

How will we acquire, process and integrate the data into CPI production?

Implementation framework overview

1) Data selection and acquisition
2) **Systems and infrastructure**
3) Data preparation and classification
4) Quality assurance and monitoring
5) **Index integration**
6) **Communication and user engagement**

Data selection and acquisition

ASSESS RELEVANCE AND COVERAGE

SECURE SUSTAINABLE DATA TRANSFER & STORAGE

MANAGE LEGAL AND CONFIDENTIALITY CONSTRAINTS

PLAN FOR CONTINUITY AND PROVIDER CHANGES

Systems and infrastructure

Challenges in modernising platforms:

Technical:

- Scale of data considerable (billions) – exponential increase in volume as more acquired
- Infrastructure & powerful compute resources needed to support production AND research
- Automation & orchestration of tools
- Secure data management & access
- Integrated front-end user interface (web-apps / dashboards)

Organisational:

- Investing in upskilling staff to increase technical skills
- Change management as data scale and approaches require adoption of new processes and tools
- Coordination within program and agency for effective use of data
- Data governance framework

StatCan – UNECE 2023

ONS – UNECE 2023

Reproducible Analytical Pipelines (RAP)

UK Government developed best practice principles for analytical systems

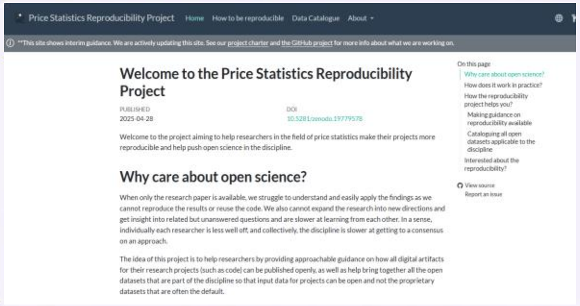
GUIDELINES AIM TO:

- Improve the quality of the analysis
- Increase trust in the analysis by producers, their managers and users
- Create a more efficient process
- Improve business continuity and knowledge management

RAP FOR CODE

- Minimise manual steps
- Use open source software
- Peer reviewed
- Uses version control
- Open sourced code and data
- Follows department good practice
- Well documented
- Tested
- Uses CI/CD
- Appropriate logging

RAP



Price Statistics Reproducibility Project Home How to be reproducible Data Catalogue About

**This site shows interim guidance. We are actively updating this site. See our project charter and the GitHub project for more info about what we are working on.

Welcome to the Price Statistics Reproducibility Project

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Welcome to the project aiming to help researchers in the field of price statistics make their projects more reproducible and help push open science in the discipline.

Why care about open science?

When only the research paper is available, we struggle to understand and easily apply the findings as we cannot reproduce the results or reuse the code. We also cannot expand the research into new directions and get insight into related but unanswered questions and are slower at learning from each other. In a sense, individually each researcher is less well off, and collectively, the discipline is slower at getting to a consensus on an approach.

The idea of this project is to help researchers by providing approachable guidance on how all digital artifacts for their research projects (such as code) can be published openly, as well as help bring together all the open datasets that are part of the discipline so that input data for projects can be open and not the proprietary datasets that are often the default.

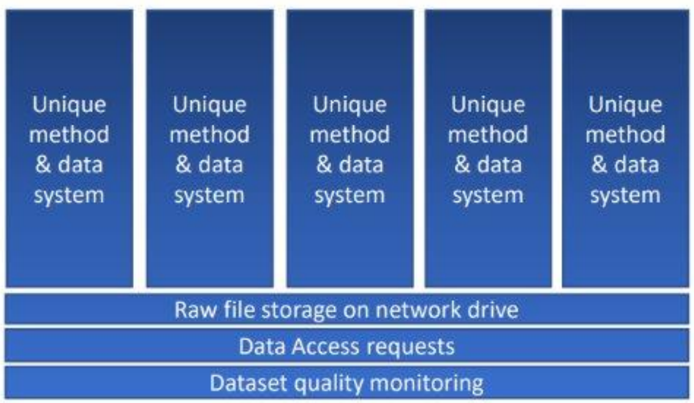
On this page
- Why care about open science?
- How does it work in practice?
- How the reproducibility project helps you?
  - Making guidance on reproducibility available
  - Cataloguing all open datasets applicable to the discipline
- Interested about the reproducibility?

View source
Report an issue

https://un-task-team-for-scanner-data.github.io/reproducibility-project/docs/

These principles help ensure that analytical systems are reliable, transparent and maintainable — supporting high quality statistics now and in the future.

Paradigm shift: from on-prem to cloud processing



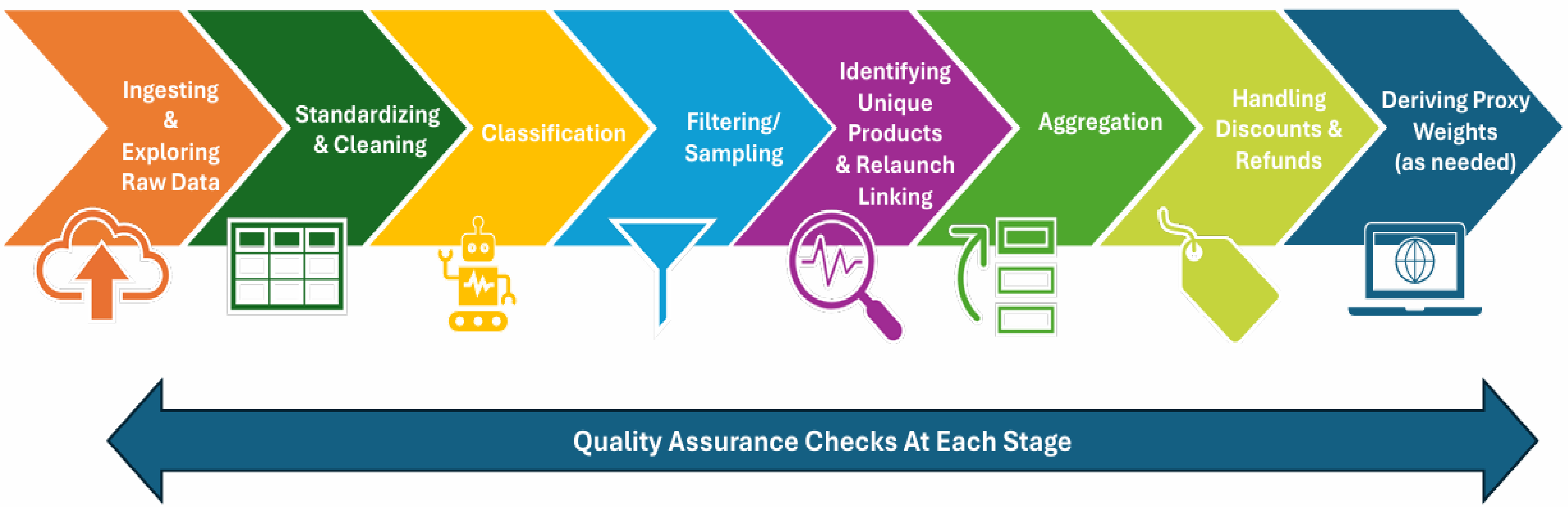
- Siloed monolithic systems (data + method specific)
- Duplicated processes
- Added complication slows down adoption and lowers transparency
- Need to reproduce what others have done

- Standardization where it makes sense
  - Standardization of common tasks for greater efficiency
- Flexible capabilities following a standard framework to support multiple outputs
- Enable adaptability to change through modularity
- Enable lighter and faster R&D

Platform environment strategy

Environment	Description	Main users	Data used	Stability
Develop	Sandbox environments where teams have full access to explore, test, and do their work.	•Software engineers •Data engineers •Infrastructure engineers	Synthetic	Not stable
Test	Test environment where all systems can work together allowing testing of individual and multiple systems.	•Testers	Synthetic	Stable
Pre-production	Environment where testing on live data can occur to ensure changes will be stable before moving into production.	•Business change team	Production (data duplicated from prod environment)	Very stable
Production	Environment where production occurs via automated scheduling of pipelines. Also, where research on data and methods can occur.	•Production team •Research team	Production (where all new data is ingested, permissions set to prevent researchers seeing “current month” data for production datasets)	Most stable

Data preparation and classification



Quality assurance across the CPI lifecycle

Quality is not a single validation stage. It is a set of control points that starts before data access and continues after publication.

01

Source assessment

Is the source suitable for statistical use?

02

Ingestion & integrity

Did the delivery arrive as expected?

03

Preparation & processing

Were products processed and classified correctly?

04

Index validation

Do the outputs behave plausibly?

05

Ongoing monitoring

Has the source or market changed?

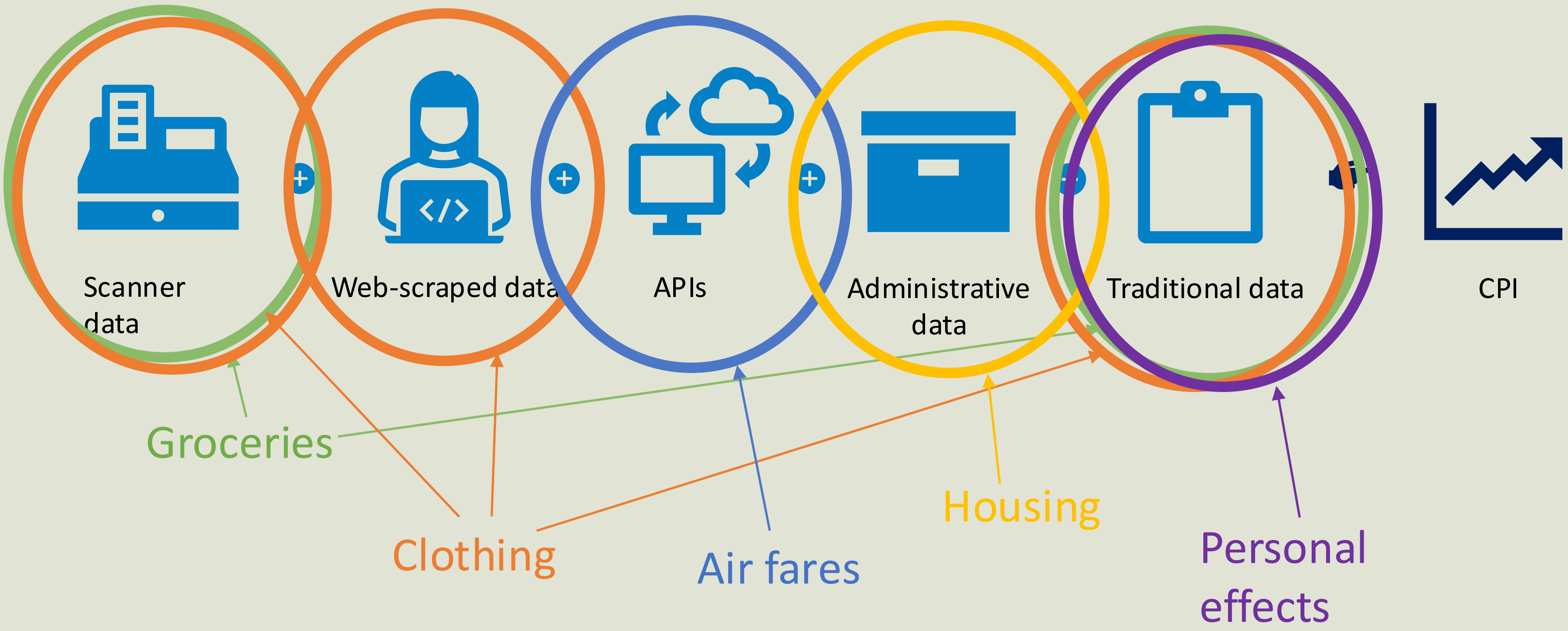
Design principle

Match the depth of assurance to the scale, complexity and operational risk of the data source.

Index integration

1. Define role within CPI (replace or supplement)
2. Manage overlap and avoid double counting
3. Align with CPI aggregation and weights
4. Maintain continuity of published series

Combining data from multiple sources



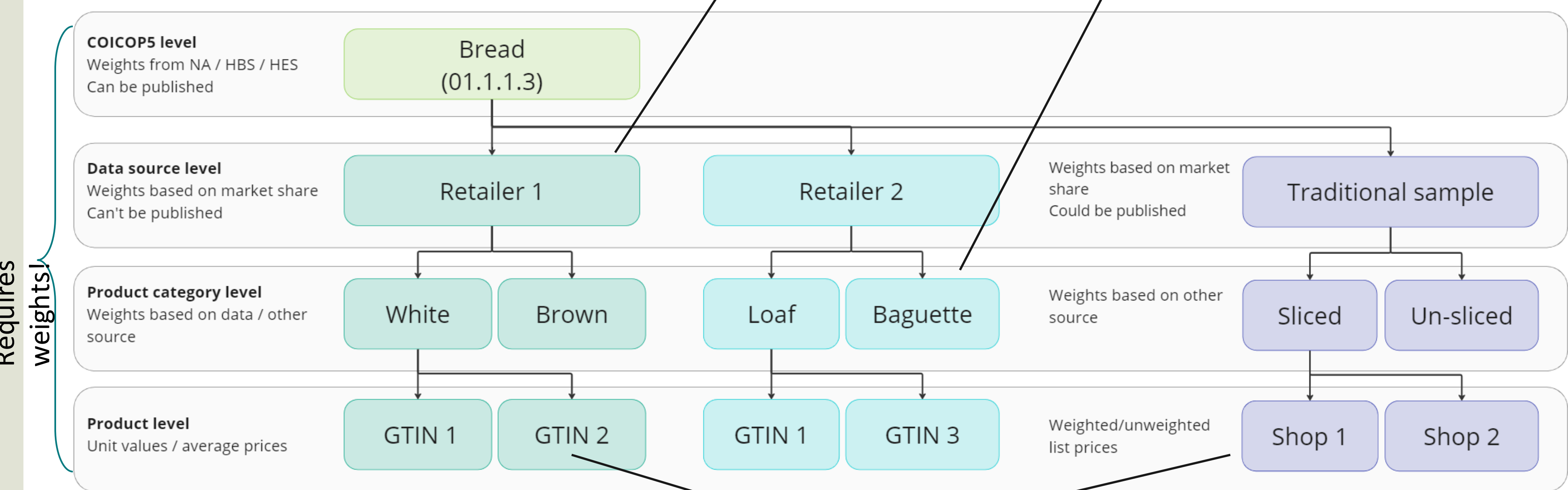
Revising structure – option 1

1. Integration at COICOP5 (sub-class)

Can't publish without disclosing retailer

Can use retailer classification

But is it accurate?



Allows for different methods at elementary aggregate level

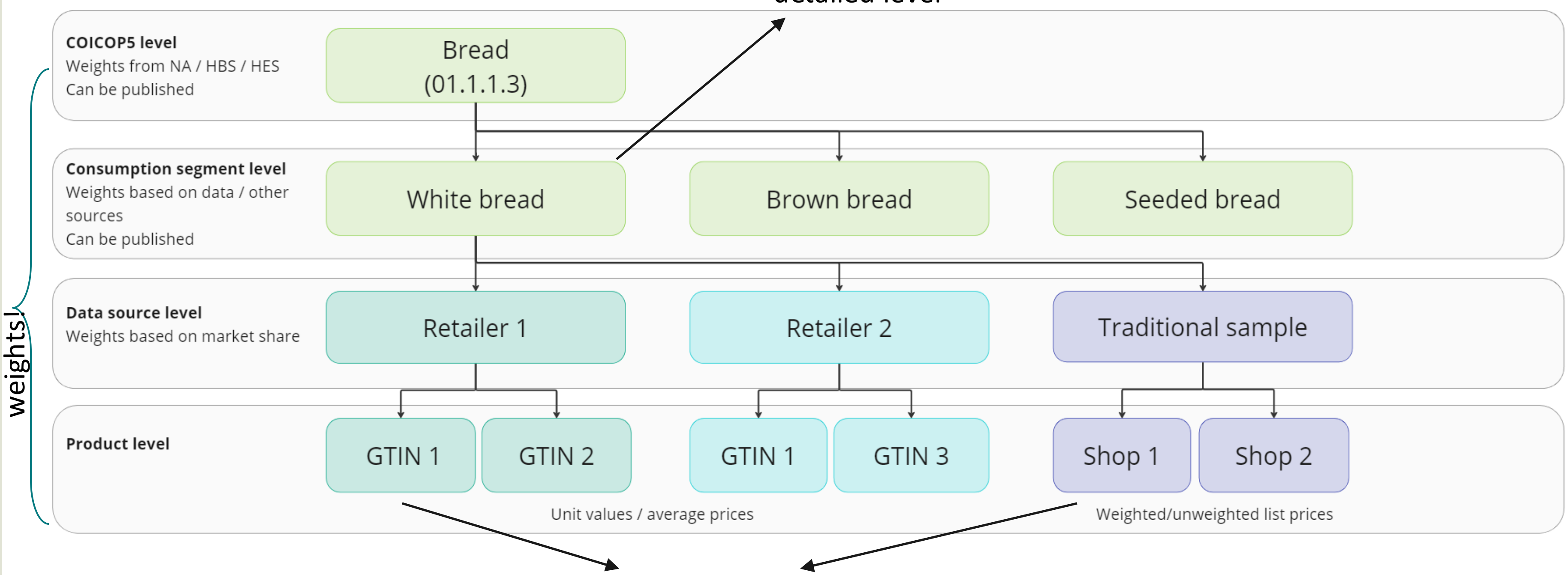
Revising structure – option 2

2. Integration at COICOP6 (consumption segment)

Can publish more detailed level

But have to classify data to this level – more challenging!

Requires weights!



Allows for different methods at elementary aggregate level

Sources of weights

- Weighting information often cannot be found from a single source and may need to utilize information from a variety of sources – a quality assessment of these sources and whether they are the best information that can be obtained, and whether they are preferable to no weighting information, should be completed.
- Examples of different sources include:
    - Household Budget/Expenditure Surveys
    - National Accounts (Household Final Consumption Expenditure)
    - Business Surveys
    - Retail Sales indicators
    - Scanner data
    - Credit card data
    - Admin data
    - Market research companies

Aggregating indices from different sources

Traditional data EA indices:

- Bilateral
- Jevons; Dutot; Lowe/Laspeyres
- Fixed base
- dec(y-1)=100
- No weights

Weights by market share / data source

Aggregation of EA indices:

- Lowe/Laspeyres
- Fixed-base
- dec(y-1)=100
- Weights from HBS / National accounts

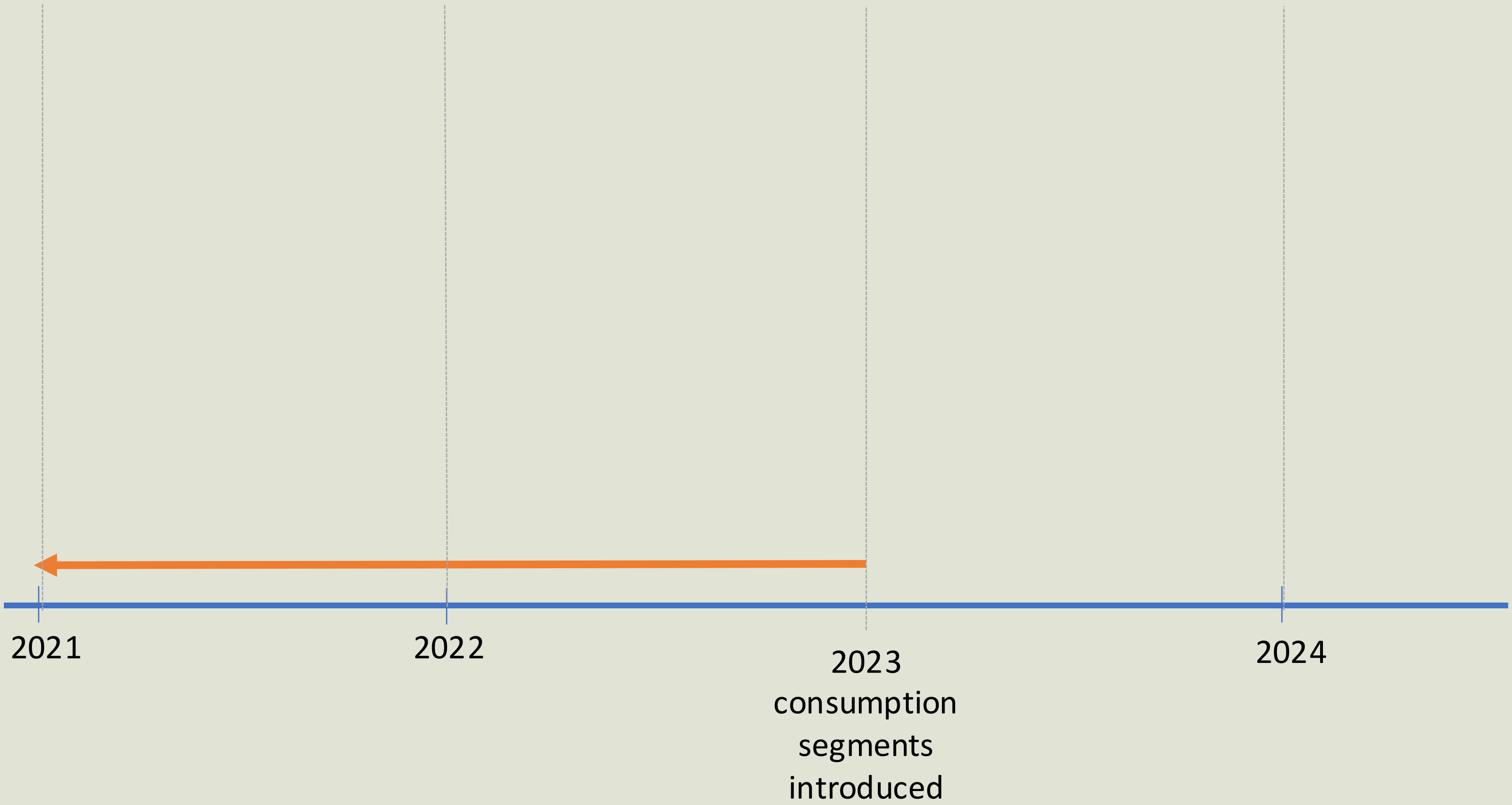
Scanner data EA indices:

- Multilateral
- GEKS-T (otherwise known as CCDI)
- Mean splice
- 25 month window, re-referenced to dec(y-1)=100
- Weights from data

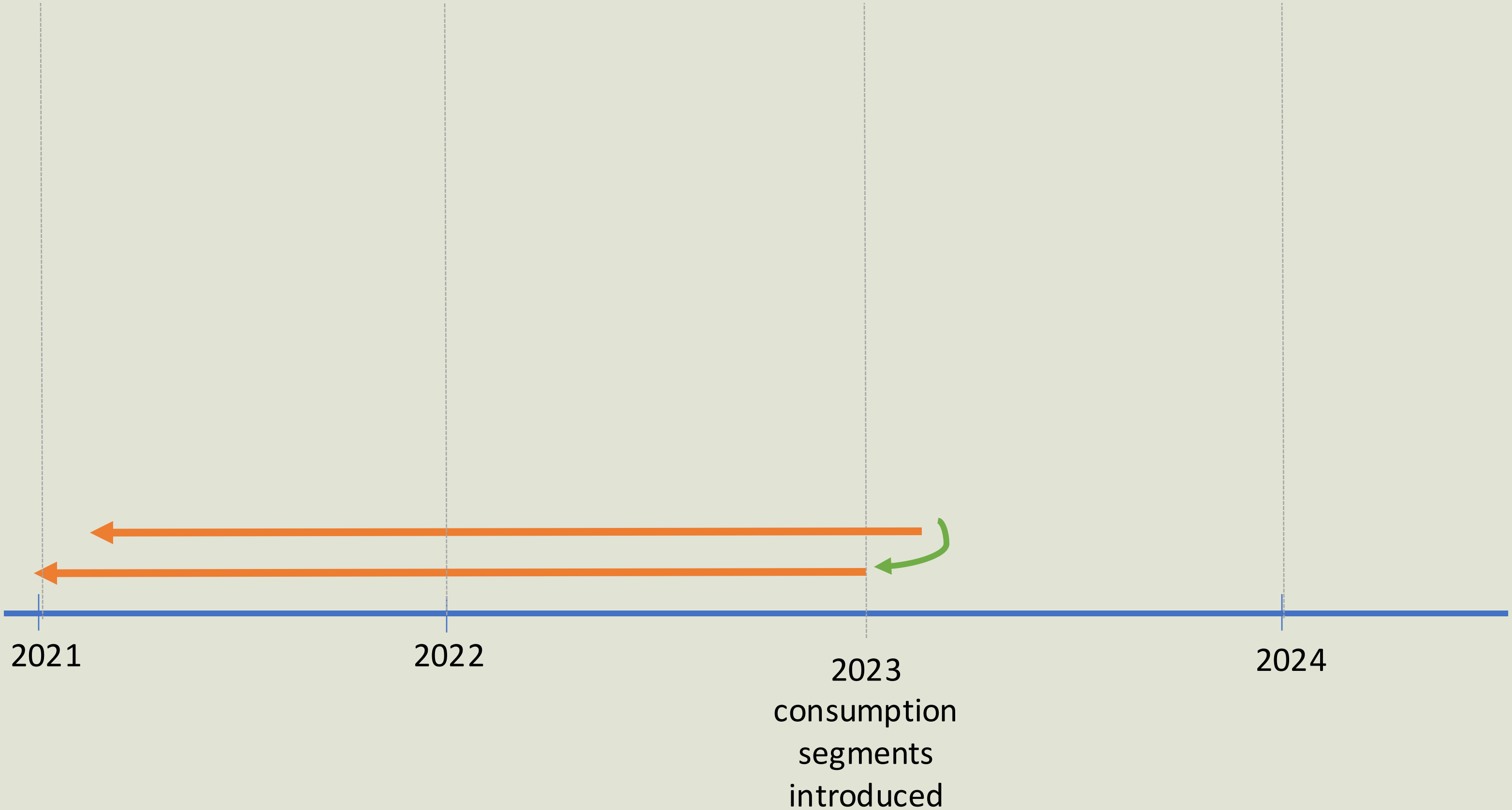
Basket updates

- New methods, data and hierarchies should only be introduced into your CPI during a basket update – through rebasing or through chainlinking
- Basket updates should be completed regularly – at least every 5 years but even more regular is better given the rapidly evolving nature of consumer spending habits and the exponential increase in access to new data sources and work to establish new methods / systems
- When working with multilateral methods with 25-month windows, it is useful to reprocess the most recent 25-months of data according to latest classifications/hierarchies etc. then splicing on subsequent months until the following basket update period where the process can be completed
- An example of this for an annually chainlinked CPI is shown in the following slides

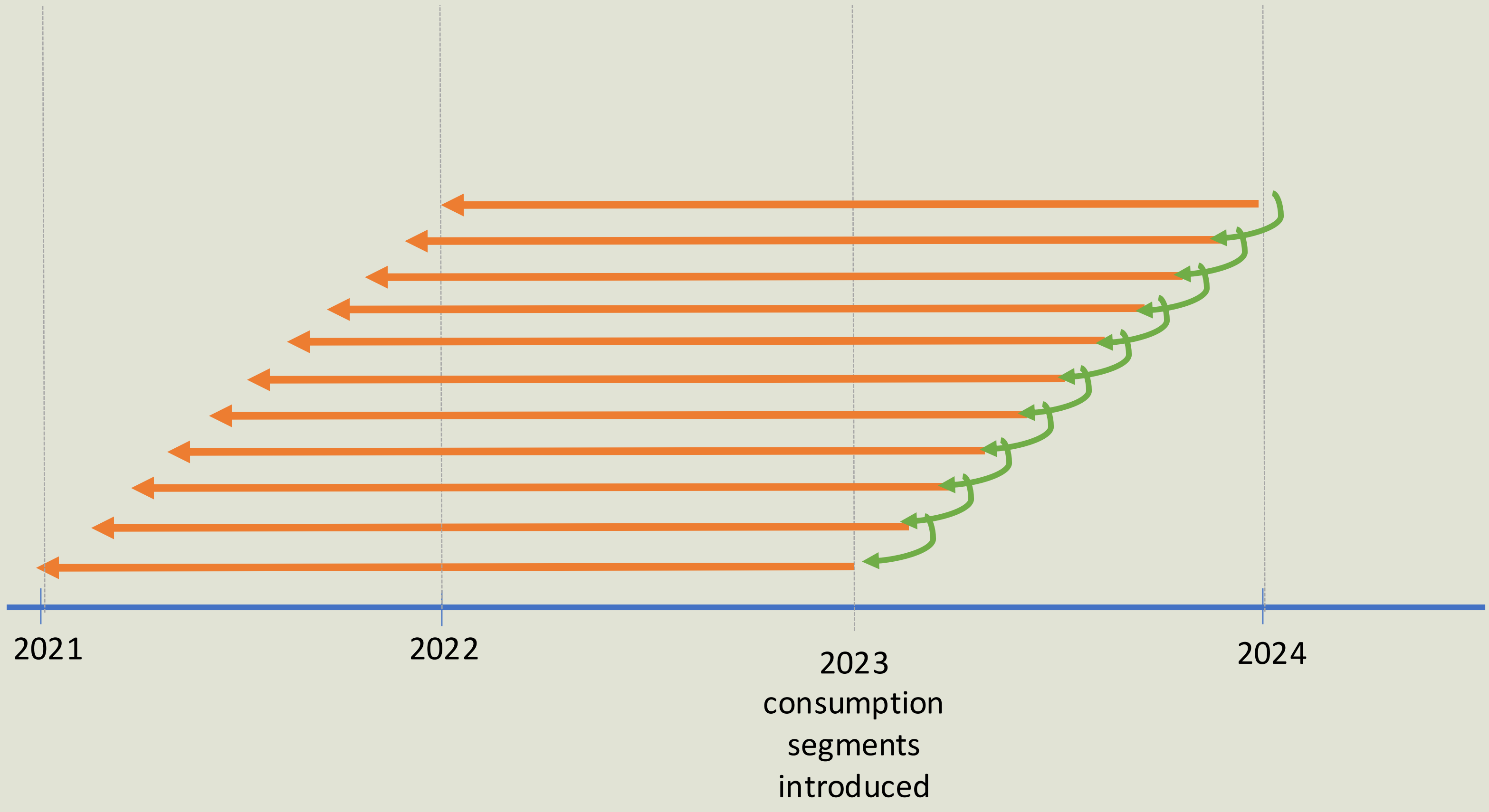
1. Clean and classify 25 months historic data and calculate GEKS across full window



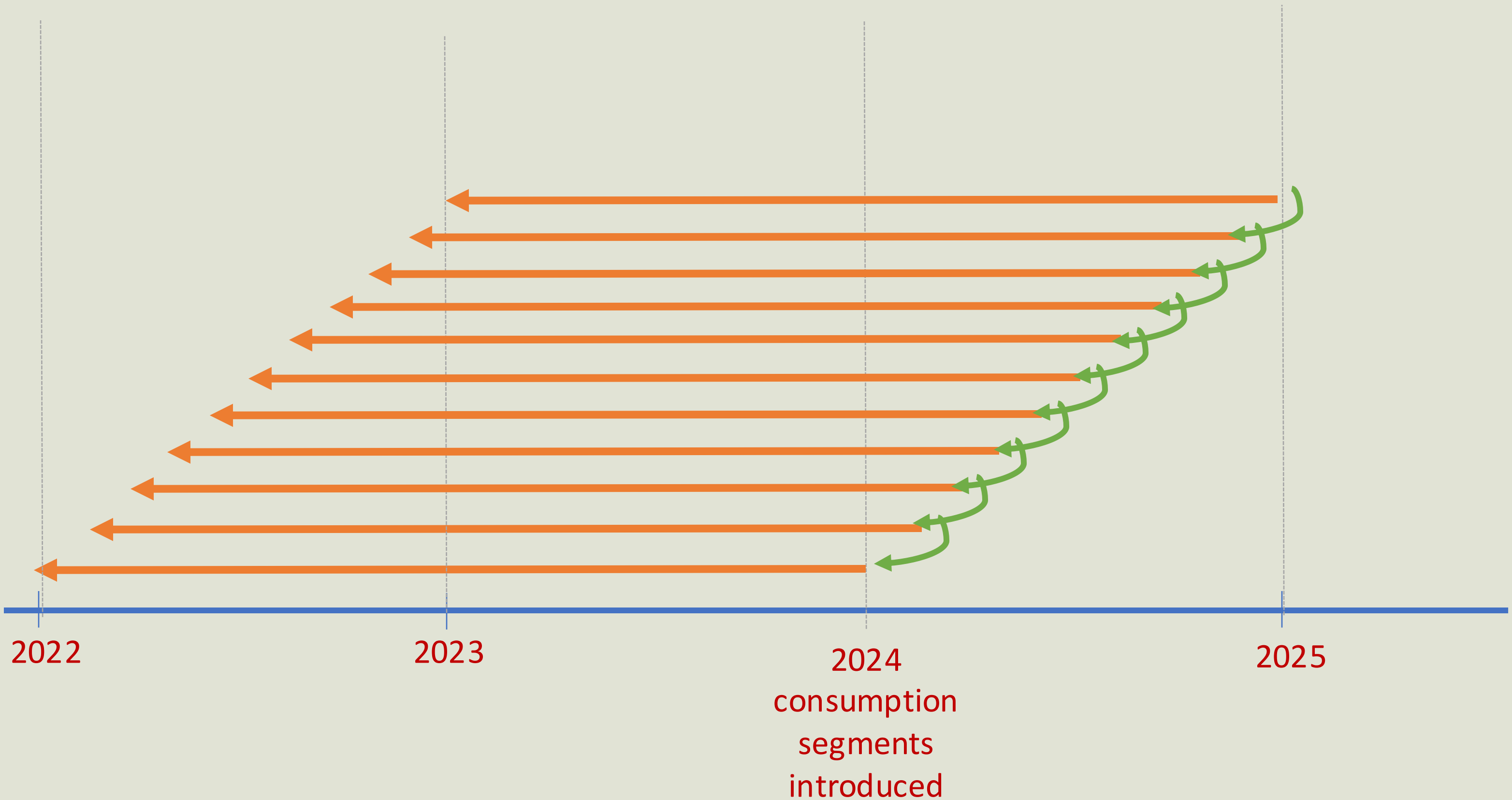
2. Append month and calculate new 25 month window, splice onto existing series, rereference to base_month(y)=100



3. Repeat monthly until new base month



4. Repeat process for each **consecutive year** with updated consumption segments



Sustain

Can we produce reliable CPI outputs over time and manage risks effectively?

Managing risks

Risk	Examples	Management
Data	- Changes in coverage, definitions, or delivery - Loss of access to data	- Monitor delivery, coverage and structure over time - Agree expectations and change protocols with providers - Maintain metadata and version tracking
Systems	- Pipeline failures or delays - Scalability and performance issues	- Use robust, modular pipelines with monitoring - Implement automated alerts and failure detection - Maintain backup and recovery procedures
Methods	- Unexpected index behaviour - Sensitivity to data changes (e.g. churn, weights)	- Conduct regular QA on outputs and contributions - Monitor sensitivity to data changes - Document methods and review periodically
Operational	- Loss of key skills or knowledge - Weak governance or unclear responsibilities	- Ensure clear roles and responsibilities - Maintain documentation and knowledge transfer - Provide training and build resilience in teams

Contingency planning

Before implementation, define potential issues, response options and operational controls.

1. WHAT HAPPENS IF DATA ARE:

- Delayed
- Incomplete
- Inconsistent
- Unavailable

Identify and assess potential data issues that could affect production.

2. COMMON CONTINGENCY OPTIONS:

- Fallback to previous data sources
- Carry forward or impute price changes
- Reduce scope (e.g. affected categories only)
- Delay publication (if necessary)
- Use alternative data sources (if available)

Define in advance the response options that can be used if issues occur.

3. OPERATIONAL CONSIDERATIONS:

- Pre-defined thresholds for action
- Clear escalation routes and decision rights
- Documented procedures and playbooks
- Regular testing of contingency plans

Put in place the controls and governance needed to respond effectively.

Contingencies should be defined before implementation and tested regularly
to help ensure continuity, minimise impact and maintain trust in CPI outputs.

Change management and communication

- Pilot studies and phased implementation
- Parallel runs & impact assessment before implementation
- Clear documentation of methods and changes and version control
- Transparent communication with users
- Clear explanation of impact on published CPI

Key messages

Integration is an institutional transformation

Requires structured and staged decisions

Must be sustainable in long-term production

Guided by CPI principles and user needs